

Exercise Sheet 11b - Solutions

Predicate Logic – Natural Deduction & Semantics

1. Here is a constructive Natural Deduction proof of $(S_1) \rightarrow \forall x. \neg 0 \leq \text{succ}(x)$

$$\begin{array}{c}
 \frac{\overline{0 \leq \text{succ}(x)}^2}{\exists x. 0 \leq x} [\exists I] \quad \frac{\overline{\neg \exists x. 0 \leq x}^1}{\neg \exists x. 0 \leq x} [\neg E] \\
 \hline
 \frac{\perp}{\neg 0 \leq \text{succ}(x)}^2 [\neg I] \\
 \frac{\neg 0 \leq \text{succ}(x)}{\forall x. \neg 0 \leq \text{succ}(x)} [\forall I] \\
 \frac{\forall x. \neg 0 \leq \text{succ}(x)}{(S_1) \rightarrow \forall x. \neg 0 \leq \text{succ}(x)}^1 [\rightarrow I]
 \end{array}$$

2. Here is a constructive Natural Deduction proof of $(S_1) \rightarrow (S_2) \rightarrow S_3$

$$\begin{array}{c}
 \frac{\overline{S_2}^2}{\forall y. 1 < y \rightarrow 0 \leq \text{succ}(y)} [\forall E] \\
 \frac{\forall y. 1 < y \rightarrow 0 \leq \text{succ}(y)}{1 < x \rightarrow 0 \leq \text{succ}(x)} [\forall E] \quad \frac{\overline{1 < x}^4}{1 < x} [\rightarrow E] \\
 \hline
 \frac{0 \leq \text{succ}(x)}{\exists x. 0 \leq x} [\exists I] \quad \frac{\overline{S_1}^1}{\exists x. 0 \leq x} [\neg E] \\
 \hline
 \frac{\exists x. 1 < x}^3 \quad \frac{\perp}{\perp}^4 [\exists E] \\
 \hline
 \frac{\perp}{S_3}^3 [\neg I] \\
 \frac{S_3}{(S_2) \rightarrow S_3}^2 [\rightarrow I] \\
 \frac{(S_2) \rightarrow S_3}{(S_1) \rightarrow (S_2) \rightarrow S_3}^1 [\rightarrow I]
 \end{array}$$

3. For example, the following models are models of $\exists x. \exists y. x \leq y \wedge \neg y < x$:

- $M_1 = \langle \mathbb{N}, \langle 0, \langle n \rangle \mapsto n + 1 \rangle, \langle \{ \langle n, m \rangle \mid n < m \}, \{ \langle n, m \rangle \mid n \leq m \} \rangle \rangle$
- and $M'_1 = \langle \mathbb{N}, \langle 0, \langle n \rangle \mapsto n \rangle, \langle \emptyset, \{ \langle n, m \rangle \mid n \leq m \} \rangle \rangle$
- and $M''_1 = \langle \mathbb{N}, \langle 0, \langle n \rangle \mapsto n \rangle, \langle \emptyset, \{ \langle n, m \rangle \mid \text{True} \} \rangle \rangle$

4. For example, the following models are models of $\neg \exists x. \exists y. x \leq y \wedge \neg y < x$:

- $M_2 = \langle \mathbb{N}, \langle 0, \langle n \rangle \mapsto n \rangle, \langle \{ \langle n, m \rangle \mid m < n \}, \{ \langle n, m \rangle \mid n < m \} \rangle \rangle$
- and $M'_2 = \langle \mathbb{N}, \langle 0, \langle n \rangle \mapsto n \rangle, \langle \{ \langle n, m \rangle \mid n < m \}, \emptyset \rangle \rangle$
- and $M''_2 = \langle \mathbb{N}, \langle 0, \langle n \rangle \mapsto n \rangle, \langle \emptyset, \emptyset \rangle \rangle$